

ROLE: you are a combination of an expert biomedical engineer, a physicist specialized in bio-electromagnetics, and a medical oncologist.

TASK: perform an exhaustive bibliographic search for a systematic review.

You must retrieve and list as many relevant original research papers as possible that match the criteria below.

Do not limit yourself to a small number; maximize the yield of relevant results within your output capacity.

RESEARCH TOPIC: the use of non-thermal and non-ionizing Electromagnetic Fields (EMFs) for cancer treatment as a stand-alone therapy.

SEARCH STRATEGY: do not limit the search to the specific acronyms listed below. Look for the broad concept of using electric or magnetic or electromagnetic fields as a therapeutic agent. Include any study that applies a physical field to treat cancer cells, regardless of the specific proprietary name or acronym used by the authors.

SEARCH QUERY PARAMETERS:

- Technological keywords: “Non-thermal electromagnetic fields” OR “Bio-electromagnetics” OR “Bioelectric therapy” OR “Electromagnetic stimulation” OR “Electric field therapy” OR “Magnetic field therapy” OR “Low intensity electric fields” OR “Intermediate frequency” OR “Radiofrequency modulation” OR “Amplitude modulated electromagnetic fields” OR “Tumor Treating Fields” OR “TTFIELDS” OR “PEMF” OR “Nanosecond Pulsed Electric Fields” OR “Pulsed Magnetic Fields” OR “Static Magnetic Fields” OR “ELF-EMF” OR “Extremely Low Frequencies Electromagnetic Fields”.
- Target keywords: “Cancer” OR “Tumor” OR “Oncology” OR “Malignancy” OR “Carcinoma”.
- Outcome keywords: “Apoptosis” OR “Proliferation” OR “Viability” OR “Necrosis” OR “Cell death” OR “Differentiation” OR “Reprogramming”.

MANDATORY FILTERS:

1. Timeframe: published between January 1, 2020, and December 2025.
2. Study type: ORIGINAL RESEARCH ONLY.
 - In vitro
 - In vivo (animal models OR human clinical trials).
 - In silico (computational simulations of field distribution/biological effect).
3. Strict exclusion: exclude reviews, systematic reviews, meta-analyses, book chapters, editorials. You must retrieve only primary data.
4. Topic exclusion: strictly exclude hyperthermia (Thermal ablation, Radiofrequency ablation), ionizing radiation (X-Ray, Gamma, Radiotherapy) and diagnostic imaging (MRI scans without therapeutic purpose).
5. Therapy exclusion: strictly exclude combinatorial therapies.
 - Do not include studies where EMF is used with chemotherapy (e.g., Doxorubicin, Paclitaxel).
 - Do not include studies using Nanoparticles (e.g., gold nanoparticles, magnetic fluid hyperthermia).
 - Do not include studies on Electroporation for drug delivery (Electrochemotherapy).

Include only papers where the EMF/Electromagnetic Field is the sole therapeutic agent tested (monotherapy).

OUTPUT FORMAT: provide a numbered list of all the papers found. For each paper, strictly follow this format:

1. Title
2. Authors
3. Year
4. Journal Name
5. DOI or direct link
6. Short summary: briefly state the specific EMF parameter used (e.g., frequency/intensity/waveform) and the main biological result.

IMPORTANT: ensure every listed paper is verifiable and continue listing until you exhaust highly relevant matches or reach your generation limit.